

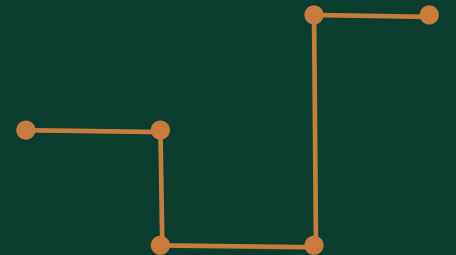
ELECTRONIC PRODUCT DESIGN AND PACKAGING

# From Silicon to System

Surface Mount Technology • SMT Components • PCB Manufacturing • PCB Design

---

*Dr. Krishnamurthy Nayak*  
*MIT Manipal*





# What We Will Cover

01

## Introduction

Packaging hierarchy: die → package → PCB → system

02

## Surface Mount Technology

Why SMT replaced through-hole assembly

03

## SMT Components

Passive chips & active packages (SOIC, QFP, BGA)

04

## Process & Reflow

Paste printing, placement, reflow profile

05

## PCB Manufacturing

From copper laminate to finished board

06

## PCB Design Rules

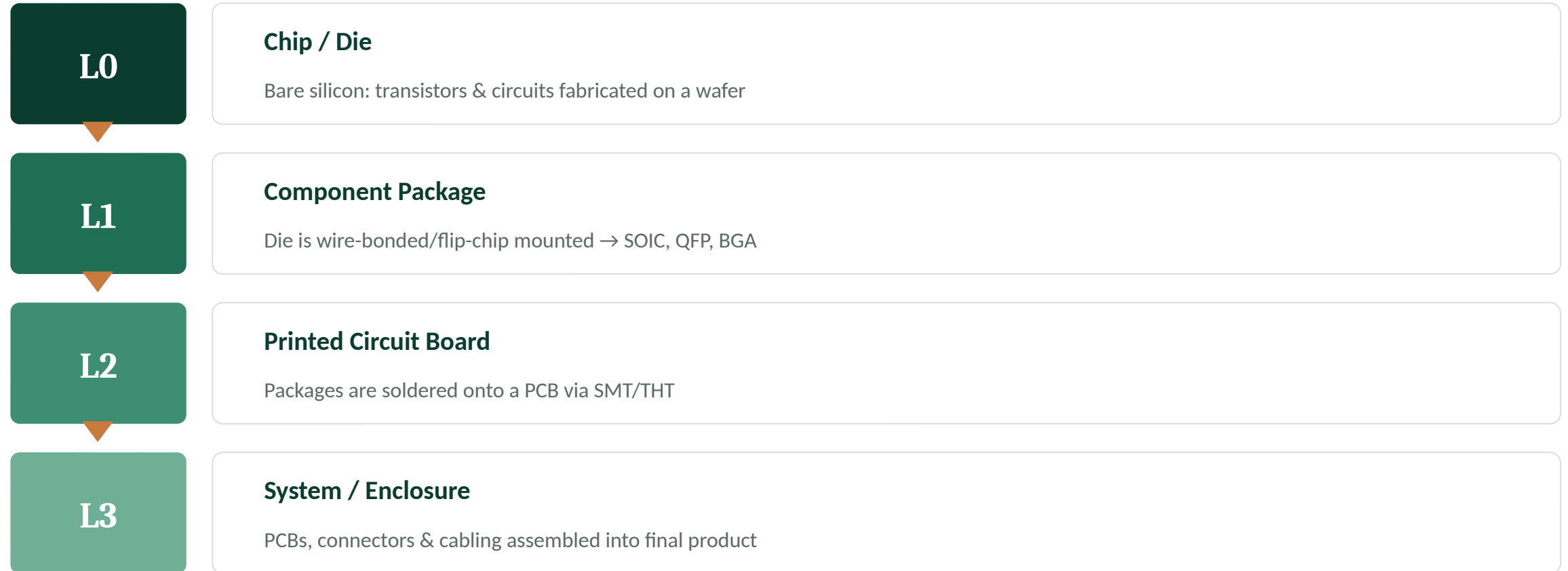
Trace sizing, stack-up, thermal & EMI rules



## INTRODUCTION

# Electronic Packaging Hierarchy

Electronic packaging interconnects and protects a bare semiconductor die, scaling it up level by level until it becomes a usable product.





# SMT vs. Through-Hole Technology

**Surface Mount Technology (SMT)** is a method of assembly where component leads are soldered directly onto pads on the PCB surface — no drilled holes required. It dominates modern electronics because of its compact size, automation-friendliness and electrical performance.

## Surface Mount (SMT)

- Components mount ON the board surface
- No drilled holes for leads
- Very small footprint (down to 0.4 x 0.2 mm)
- Fully automated pick-and-place + reflow
- High component density, shorter leads → better RF/HF performance
- Used in phones, laptops, wearables

## Through-Hole (THT)

- Leads inserted through drilled holes
- Soldered on the opposite (bottom) side
- Larger footprint, stronger mechanical bond
- Manual or wave soldering
- Preferred for connectors & high-power / high-stress parts
- Used in power supplies, connectors, relays



# The SMT Process Flow



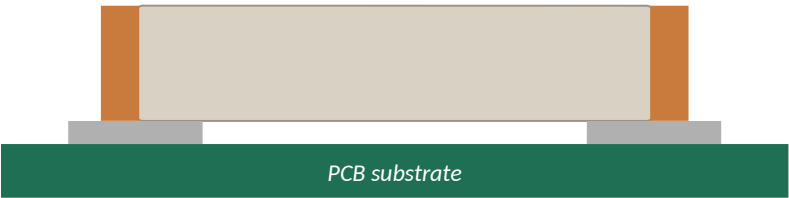
**Why reflow, not wave soldering for SMT?** Reflow melts pre-deposited paste in a controlled thermal profile so tiny, closely-spaced pads solder reliably without bridging — wave soldering (molten solder wave) is reserved for through-hole leads.



# Passive Components & Chip Sizes

## Chip Resistor / Capacitor — Cross Section

Ceramic / resistive body



Solder termination (copper/tin)

## Common Chip Sizes (Imperial → Metric)

| 0201 | 0.60 x 0.30 | High-density mobile boards    |
|------|-------------|-------------------------------|
| 0402 | 1.00 x 0.50 | Smartphones, wearables        |
| 0603 | 1.60 x 0.80 | General-purpose SMT           |
| 0805 | 2.00 x 1.25 | Power-tolerant, easier rework |
| 1206 | 3.20 x 1.60 | Higher power resistors/caps   |

*Naming convention: digits = dimensions in hundredths of an inch (e.g., 0603 = 0.06" x 0.03").*



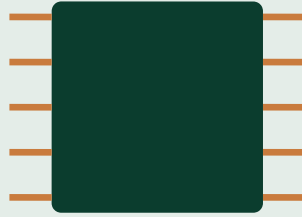
# Active Device Packages



## SOIC

*Small Outline IC*

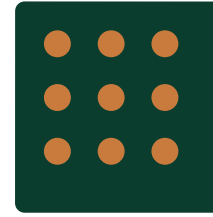
Gull-wing leads on 2 sides, 1.27 mm pitch — op-amps, logic ICs



## QFP

*Quad Flat Package*

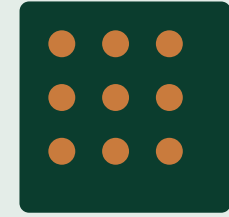
Gull-wing leads on all 4 sides, 0.4–0.8 mm pitch — MCUs, FPGAs



## BGA

*Ball Grid Array*

Solder balls in a grid under the package — high pin-count SoCs



## QFN

*Quad Flat No-lead*

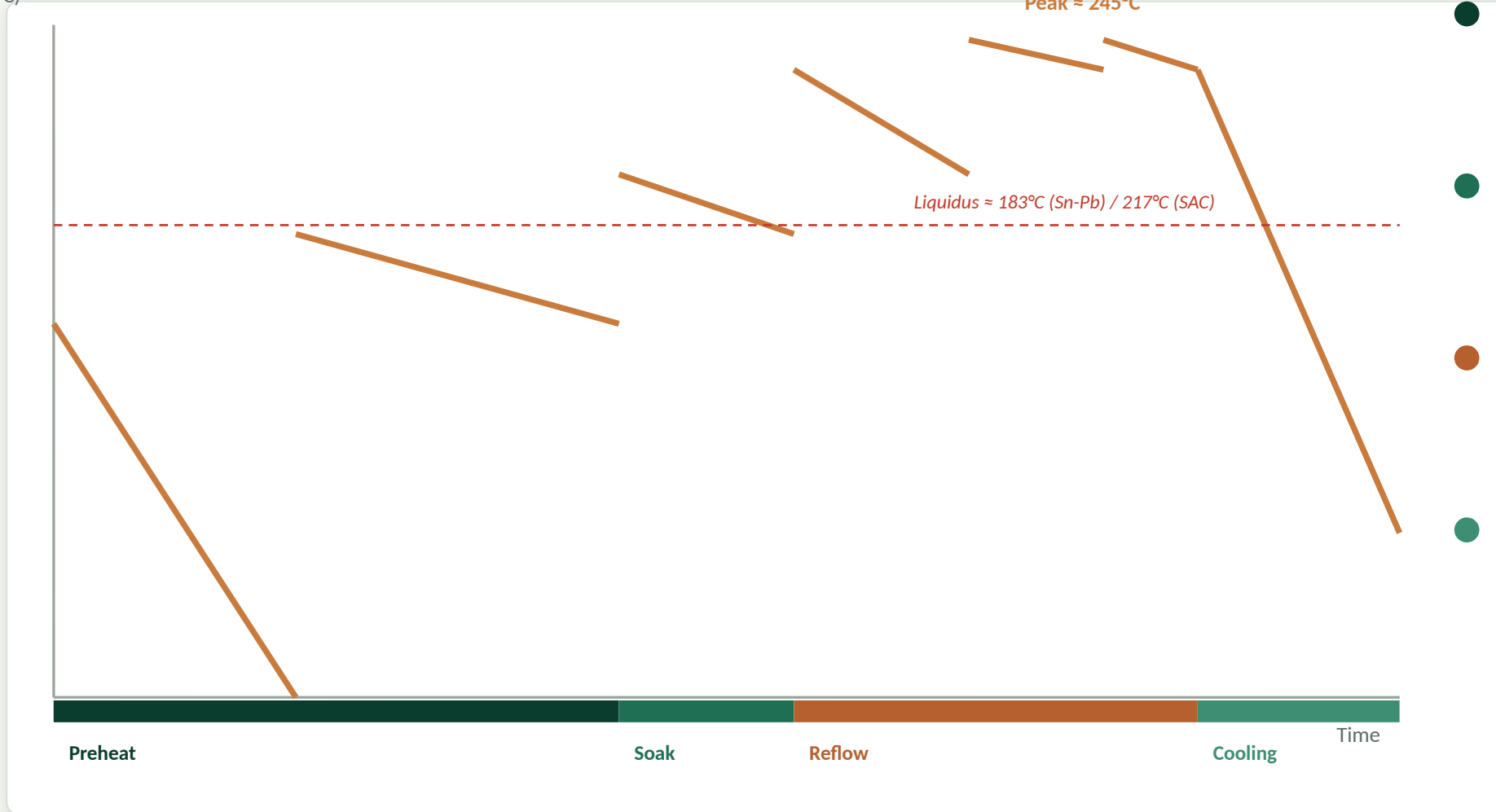
Pads flush with the package bottom — compact, good thermal path



SMT PROCESS

# The Reflow Soldering Profile

Temp (°C)



## Preheat

Ramp ≤ 3°C/s to avoid thermal shock

## Soak

Activates flux, equalises board temperature

## Reflow

Above liquidus for 30–60 s — joints form

## Cooling

Controlled cool-down → fine, strong grain



# Solder Paste Volume via Stencil Printing

**Problem:** A rectangular stencil aperture measures 1.0 mm x 0.5 mm for a 0402 chip pad. Stencil thickness = 0.12 mm. Find the solder paste volume deposited per pad, and the paste weight if density = 7.4 g/cm<sup>3</sup>.

Step 1 — Aperture area

$$A = L \times W = 1.0 \text{ mm} \times 0.5 \text{ mm} = 0.50 \text{ mm}^2$$

Step 2 — Paste volume

$$V = A \times t = 0.50 \text{ mm}^2 \times 0.12 \text{ mm} = 0.060 \text{ mm}^3$$

Step 3 — Convert to cm<sup>3</sup>

$$V = 0.060 \text{ mm}^3 = 0.00006 \text{ cm}^3 \quad (1 \text{ cm}^3 = 1000 \text{ mm}^3)$$

Step 4 — Paste mass

$$m = \rho \times V = 7.4 \text{ g/cm}^3 \times 0.00006 \text{ cm}^3 = 4.44 \times 10^{-4} \text{ g} \quad (\approx 0.44 \text{ mg})$$

**Result:**  $\approx 0.06 \text{ mm}^3$  of paste ( $\sim 0.44 \text{ mg}$ ) is deposited per 0402 pad — stencil aperture area and thickness directly set the solder joint volume.



# PCB Materials & Multilayer Stack-up

|                           |
|---------------------------|
| Silkscreen                |
| Solder Mask               |
| Copper (Top) - L1         |
| FR-4 Prepreg (dielectric) |
| Copper (Inner) - L2 GND   |
| FR-4 Core                 |
| Copper (Inner) - L3 PWR   |
| FR-4 Prepreg (dielectric) |
| Copper (Bottom) - L4      |
| Solder Mask               |

## Core Substrate

FR-4: woven glass fabric + epoxy resin. Rigid, low-cost, good insulator (dielectric constant  $\approx 4.5$ ).

## Copper Layers

Standard weight: 1 oz/ft<sup>2</sup>  $\approx$  35  $\mu$ m thick. Etched to form traces, planes & pads.

## Solder Mask & Silkscreen

Mask (usually green) insulates copper except pads; silkscreen prints labels & reference designators.

## Layer Count

2-layer: simple circuits. 4-6 layer: dedicated GND/PWR planes. 8+ layers: dense, high-speed digital designs.



# From Copper Laminate to Finished Board



## Two Core Techniques

**Subtractive process:** start with fully copper-clad laminate, etch away the unwanted copper to leave traces (most common method).

**Additive / semi-additive:** copper is deposited only where traces are needed — used for very fine-pitch, high-density boards.

# Trace Width from Current-Carrying Capacity

## IPC-2221 External Trace Formula

$$I = k \cdot \Delta T^{0.44} \cdot A^{0.725}$$

$I$  = max current (A)  $k$  = 0.048 (external) / 0.024 (internal)  $\Delta T$  = allowed temperature rise (°C)  $A$  = cross-section area (mils<sup>2</sup>)

**Given:** External trace,  $I = 2$  A,  $\Delta T = 20^\circ\text{C}$ , copper thickness = 1 oz/ft<sup>2</sup> (1.37 mils). Find the minimum trace width.

$$\rightarrow A = [ I / (k \cdot \Delta T^{0.44}) ]^{(1/0.725)}$$

$$\rightarrow A = [ 2 / (0.048 \times 20^{0.44}) ]^{1.379}$$

$$\rightarrow A = [ 2 / 0.171 ]^{1.379} = (11.7)^{1.379} \approx 25.9 \text{ mils}^2$$

$$\rightarrow \text{Width} = A / \text{thickness} = 25.9 / 1.37 \approx 18.9 \text{ mils} \approx 0.48 \text{ mm}$$

## Typical PCB Design Rules

Min. trace width 3–4 mil (fine-pitch); 8 mil (standard)

Min. trace spacing 3–4 mil (fine-pitch); 8 mil (standard)

Min. via drill / pad 8 mil / 16 mil (typical)

Copper-to-edge clearance  $\geq 10$  mil (20 mil safer)

Annular ring  $\geq 4$  mil around drill hole

Result (this example)  $\approx 19$  mil trace — well above minimum



# Key Design Considerations

1

## Thermal Management

Place hot components near board edge/vents; use thermal vias & copper pours to spread heat; keep sensitive parts away from heat sources.

2

## Signal Integrity & EMI

Keep high-speed traces short with controlled impedance; use ground planes; avoid sharp 90° trace corners; separate analog/digital grounds.

3

## Manufacturability (DFM)

Respect min. trace/space/via rules; keep uniform component orientation; avoid components too close to board edge or each other.

4

## Power Distribution

Use wide traces or planes for power/ground; decouple ICs with local bypass capacitors placed close to supply pins.



## PRACTICE

# Practice Questions

Q1

Differentiate between SMT and THT assembly in terms of density, cost and mechanical strength.

Q2

Sketch and label the four zones of a typical reflow soldering temperature profile.

Q3

A stencil aperture is 0.8 mm x 0.4 mm with 0.15 mm thickness. Calculate the solder paste volume deposited.

Q4

List and explain any four PCB design rules that must be followed for manufacturability (DFM).

Q5

Using IPC-2221, find the trace width needed to carry 1.5 A on an internal layer with  $\Delta T = 10^\circ\text{C}$ .

Q6

Why are ground/power planes used in multilayer PCBs? How do they help EMI and power distribution?



RECAP

# Key Takeaways

1

## Packaging hierarchy

Die → package → PCB → system, each level trading off size, cost & performance.

2

## SMT dominates

Smaller footprint, automation-friendly, better high-frequency performance than THT.

3

## Component families

Passives (chip R/C by size code) and active packages (SOIC, QFP, BGA, QFN).

4

## Reflow profile

Preheat → Soak → Reflow → Cool — controls joint quality & reliability.

5

## PCB manufacture

Subtractive etching + drilling + plating + masking builds the copper circuit.

6

## Design rules matter

Trace width (IPC-2221), spacing, thermal & EMI rules ensure a manufacturable, reliable board.